

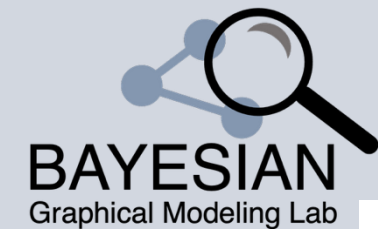
# Bayesian Graphical Modeling

## Topological Priors

*What is psychometrics? 2025*

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[bayesiangraphicalmodeling.com](http://bayesiangraphicalmodeling.com)



# Session setup



Part I: Theory



Part II: Tutorial / Practical



# Session setup



## Part I: Theory

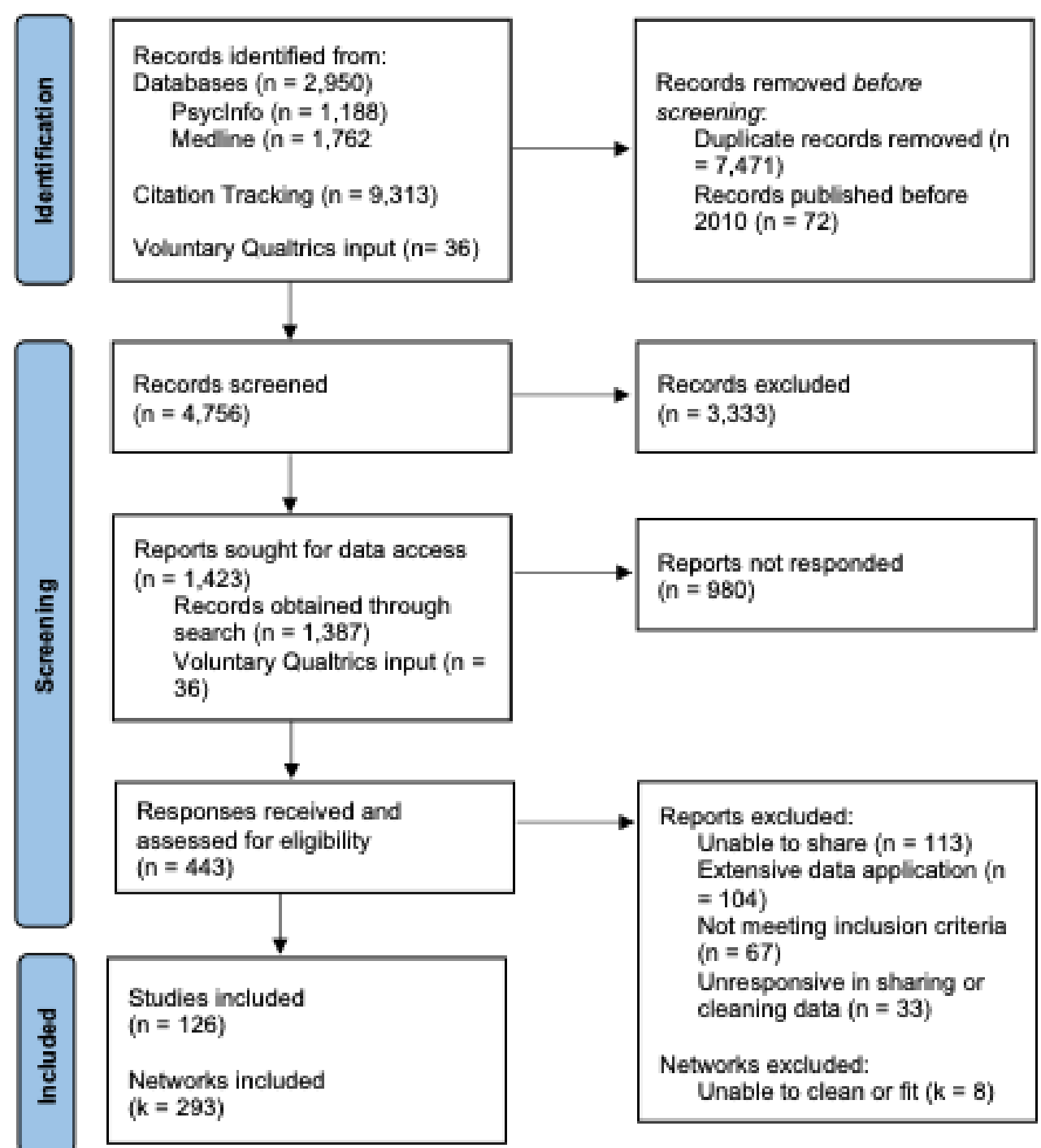
# Whats the evidence?

Contrary to most existing network estimation methods, the **Bayes factor analyses** in Session I allowed us to ***quantify evidence of absence***, not just detect ***absence of evidence***.

This naturally raises a question: **What happens when we apply this Bayesian reasoning to existing, published networks?**

To explore this, we performed a **Bayesian reanalysis** of previously reported networks.

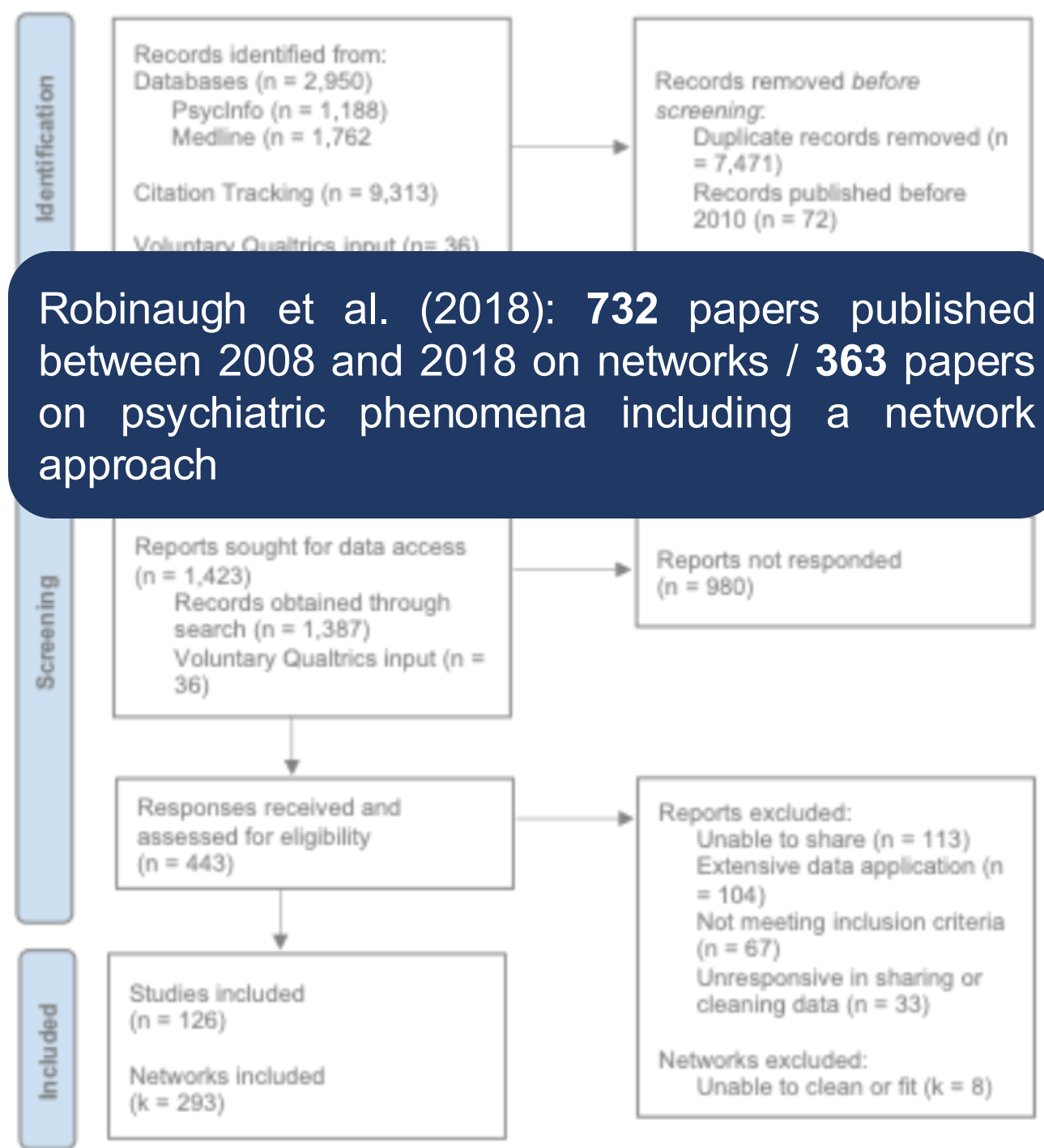
# Papers and data



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Screening 4,720 papers

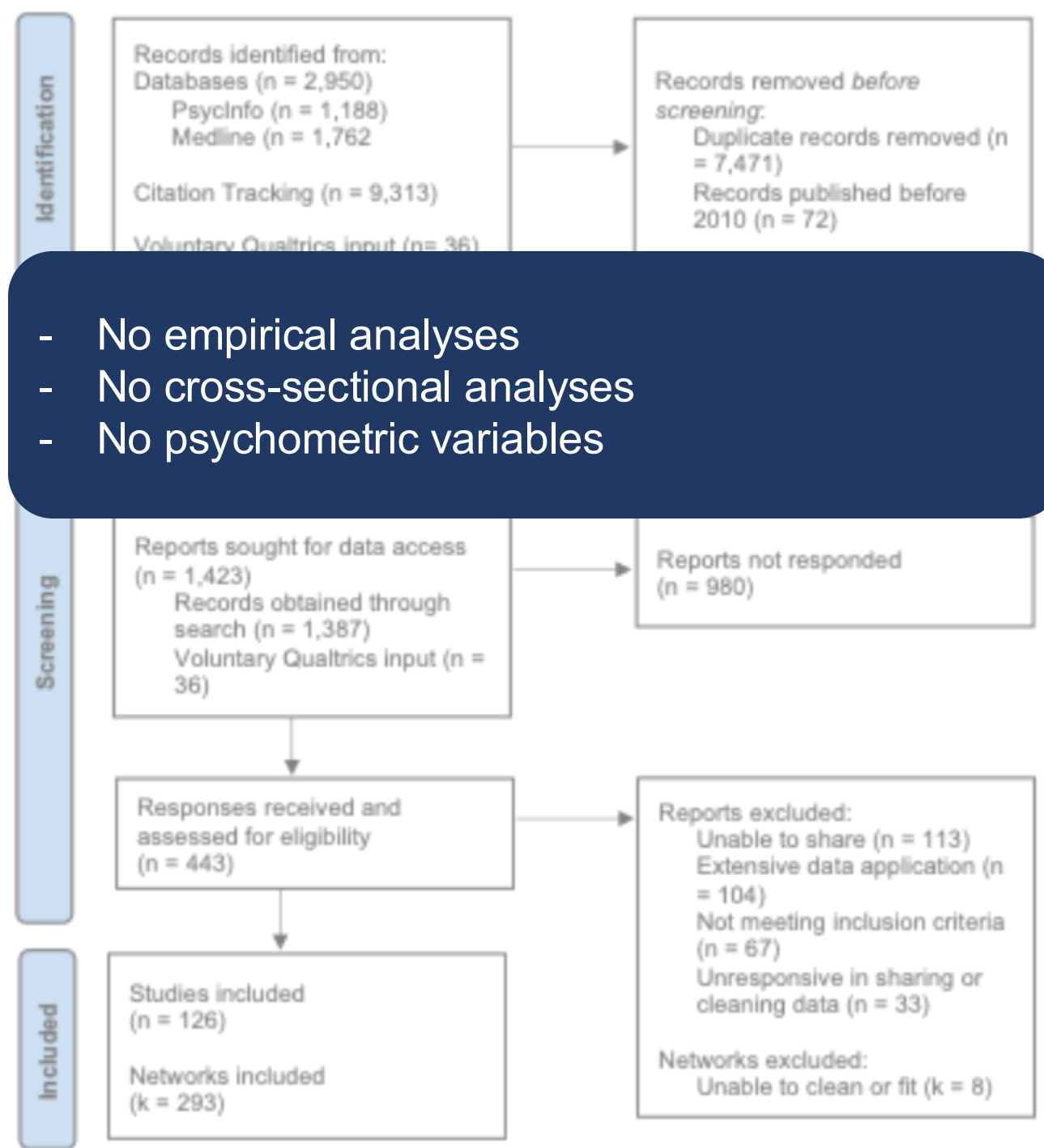
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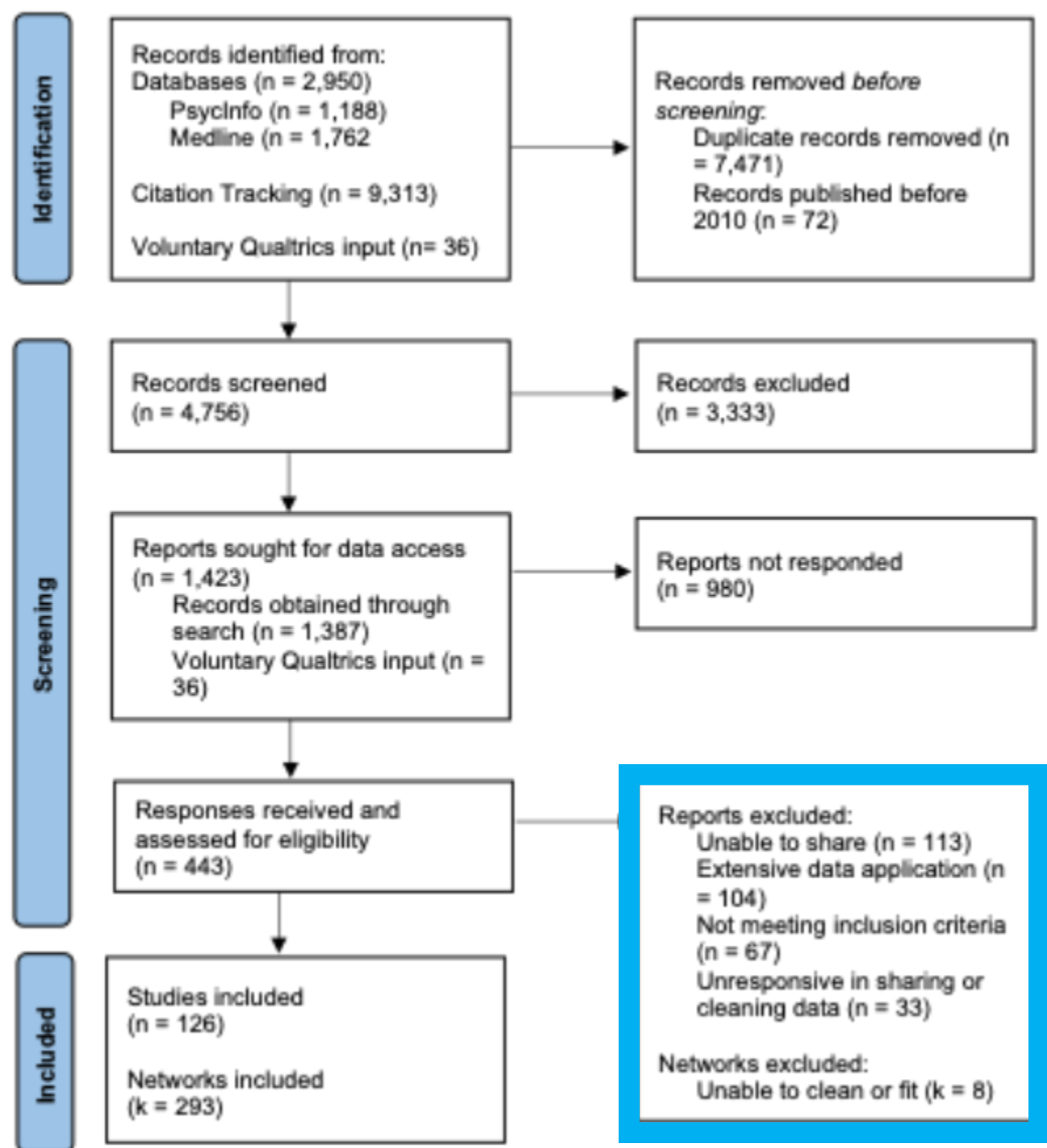
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- 443 responses
  - 126 included
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→ 126 papers; 293 networks





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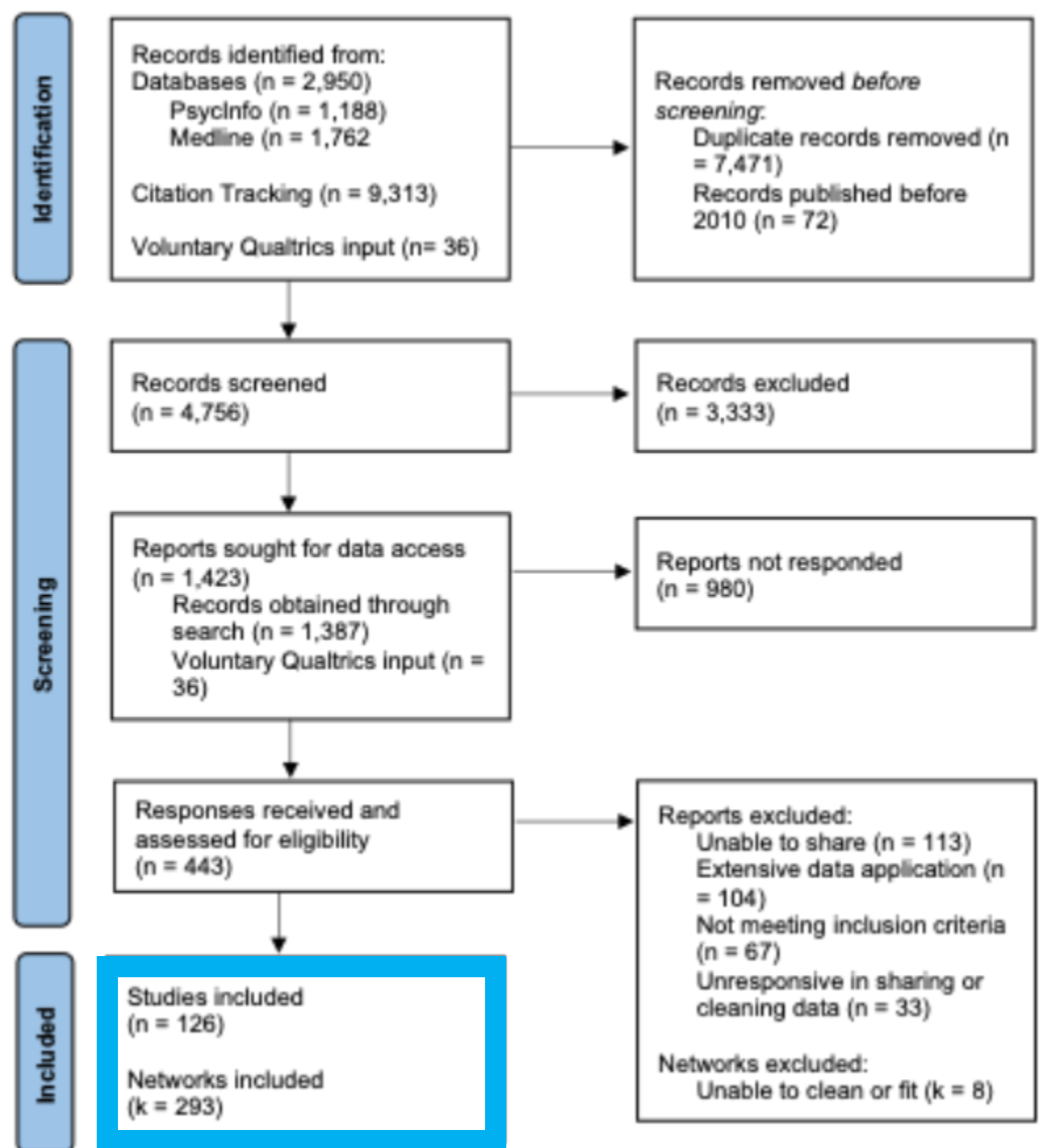
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# Analysis

We analyzed all networks using the Gaussian graphical model

For each network relation, we use the Bayes factor to assess the support in the data for the competing hypotheses:

$\mathcal{H}_0$ : Partial association is zero

$\mathcal{H}_1$ : Partial association is non-zero

# Analysis

The Bayes factor indicates how much more plausible the data are under one hypothesis relative to the other

$$\text{BF}_{10} = \frac{p(\text{data} \mid \mathcal{H}_1)}{p(\text{data} \mid \mathcal{H}_0)}$$

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Evidence for  $\mathcal{H}_0$

Evidence for  $\mathcal{H}_1$

$$\text{BF}_{10} = \frac{1}{30}$$

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$$\text{BF}_{10} = \frac{1}{3}$$

$$\text{BF}_{10} = 1$$

$$\text{BF}_{10} = 3$$

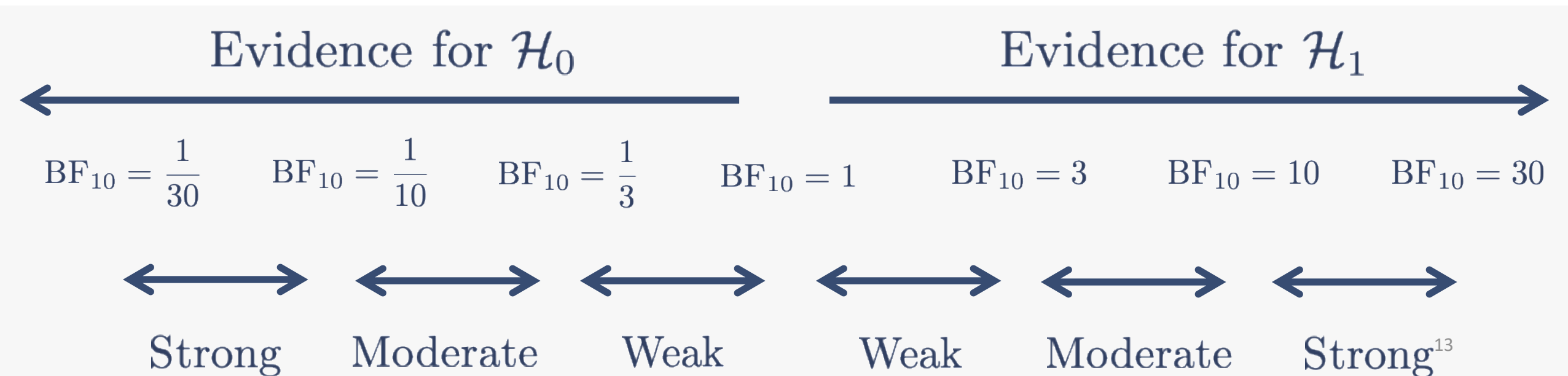
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$$\text{BF}_{10} = 30$$

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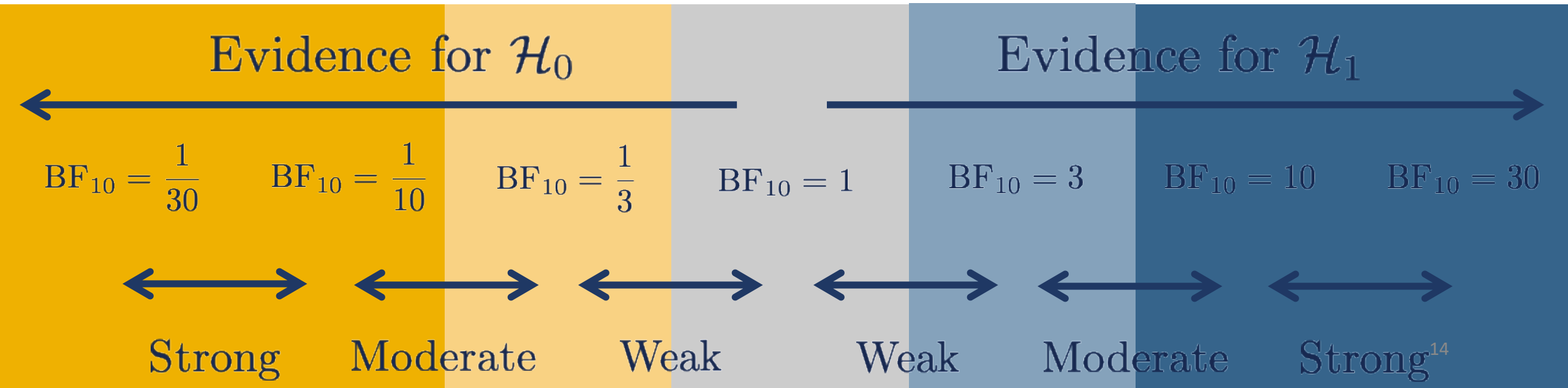
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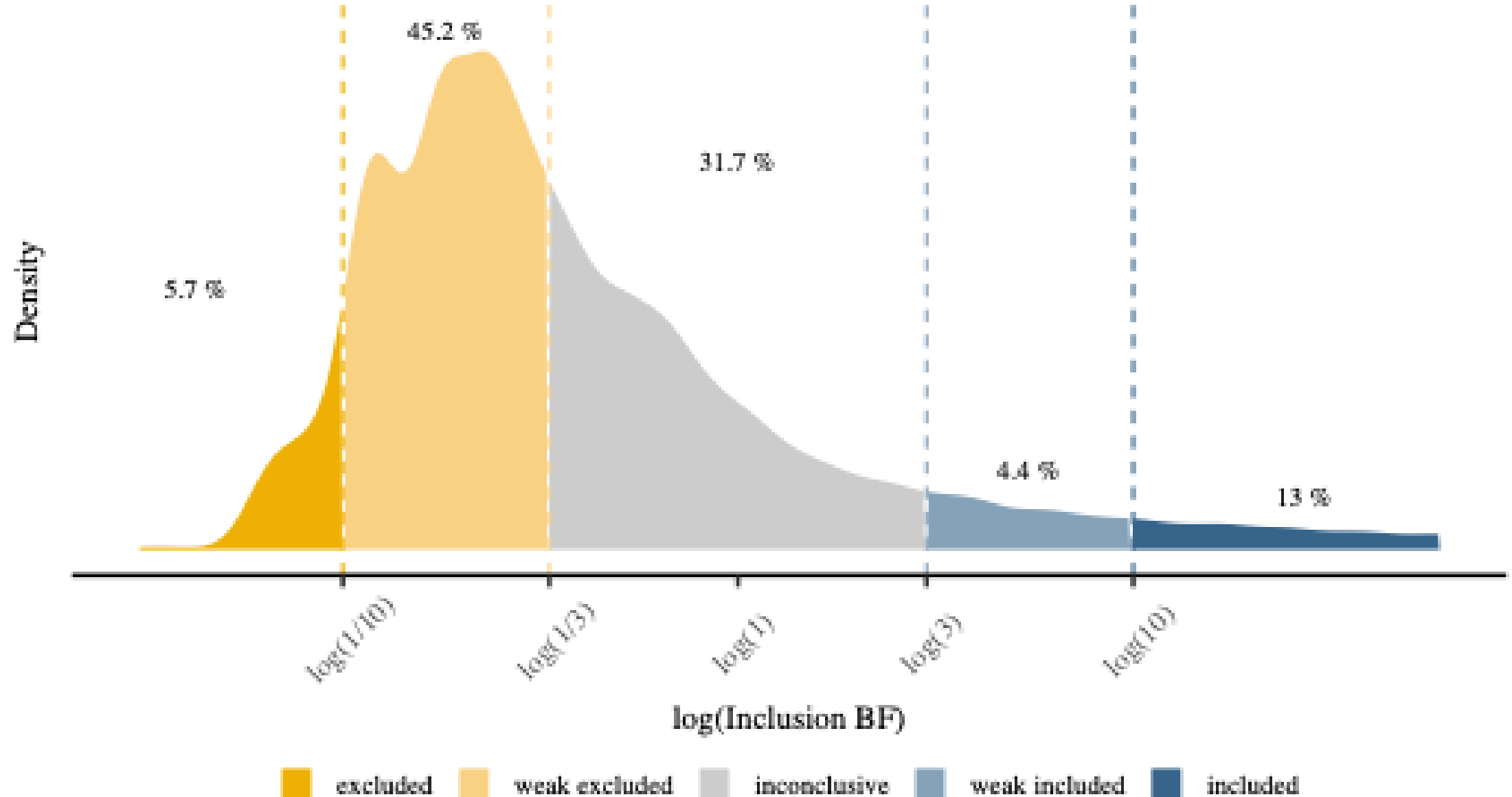
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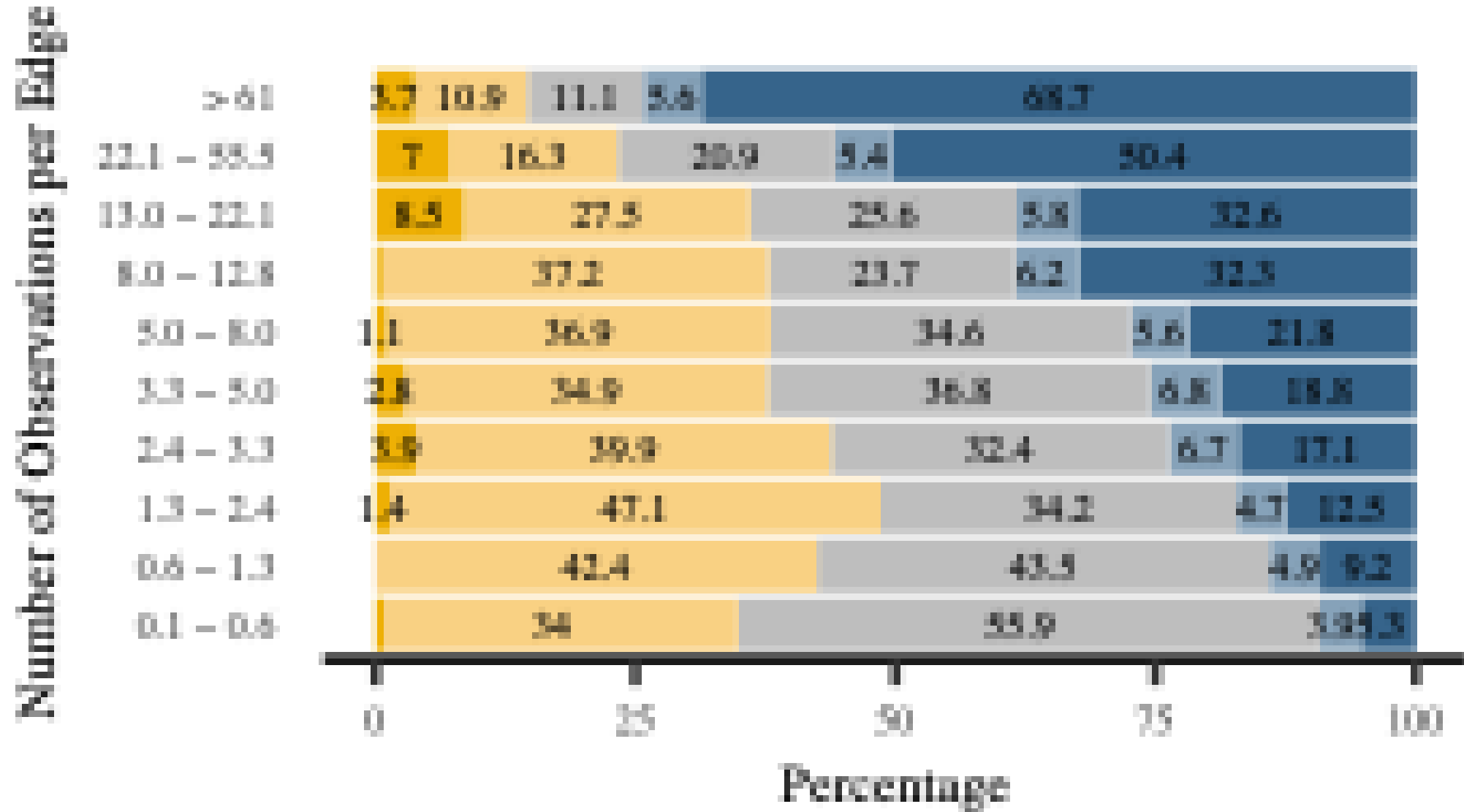
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# Results



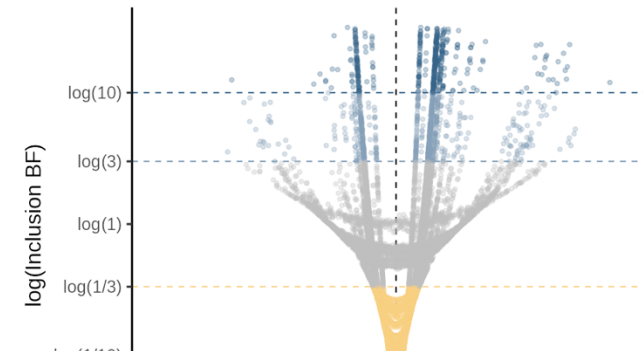
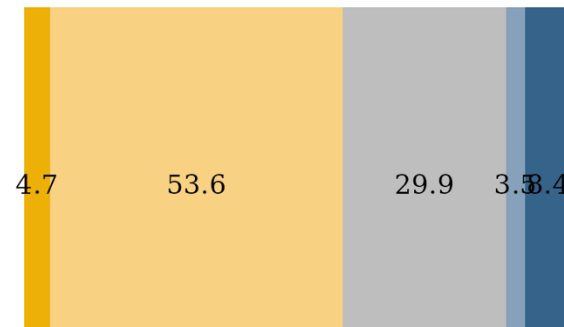
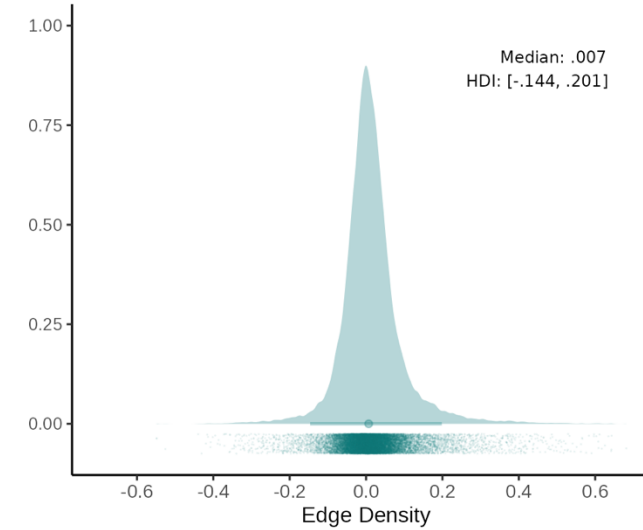
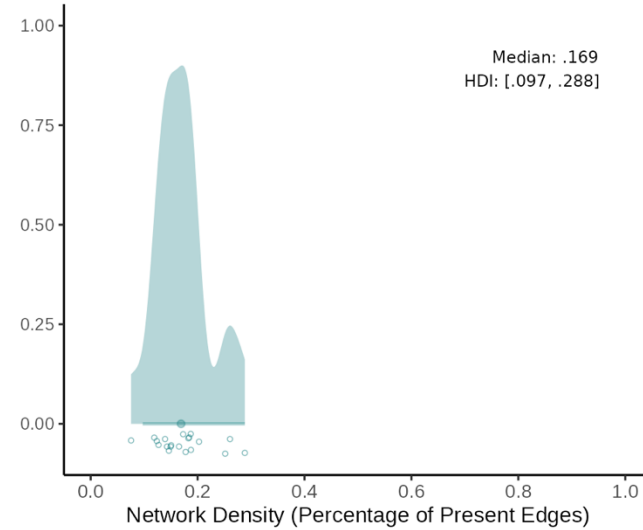
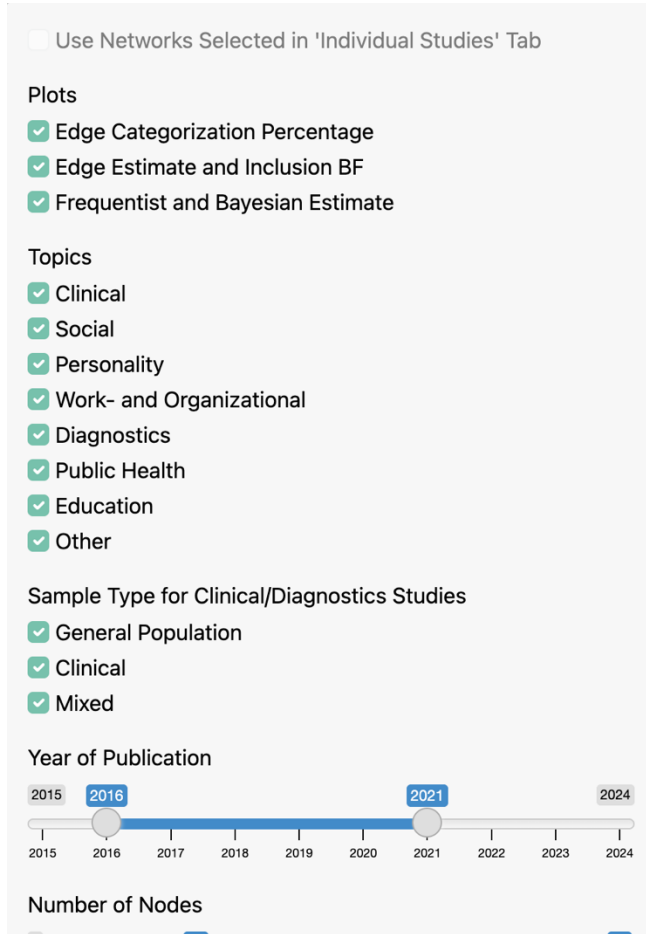
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<https://uvasobe.shinyapps.io/ReBayesed/>



# Conclusions

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3. Inconclusive evidence of remaining ~30% of edges  
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4. Strength of evidence strongly determined by (relative) sample size  
Median was 5 obs. per edge ( $1 < , > 7,000$ ).

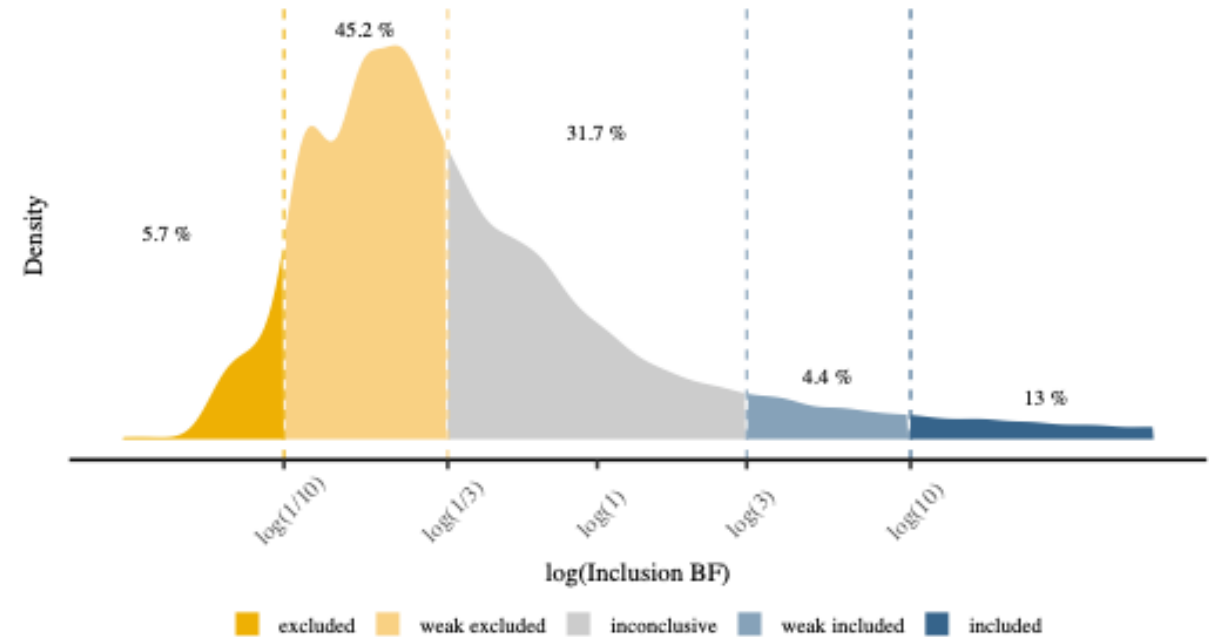
# Limitations

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We did not compare to original conclusions

- Might have indicated uncertainty

- Might not have been interested in the structure/edges

# What did the reanalysis reveal?

Bayes factors for most edges were **small** → evidence for inclusion or exclusion often **inconclusive**

As a result, there is **considerable uncertainty** about the overall network structure

Our Bayes factor methods **ignored multiplicity** → overstate confidence in individual edges

Together, this shows the need to **model structural uncertainty explicitly**

# Existing models/priors for network structures

## The independent Bernoulli model

$$\begin{aligned} p(\gamma_{12}, \dots, \gamma_{(p-1)p}) &= \prod_{i=1}^{p-1} \prod_{j=i+1}^p \pi^{\gamma_{ij}} (1 - \pi)^{1-\gamma_{ij}} \\ &= \pi^k (1 - \pi)^{m-k} \end{aligned}$$

Here,  $k$  is number of active edges,  $m$  total number of edges. But what is the inclusion probability?

Marsman, M., Huth, K., Waldorp, L., & Ntzoufras, I. (2022). Objective Bayesian Edge Screening and Structure Selection for Ising Networks. *Psychometrika*, 87, 47-82.

Mohammadi, R., & Wit, E. (2019). BDgraph: An R Package for Bayesian Structure Learning in Graphical Models. *Journal of Statistical Software*, 89, 1–30.

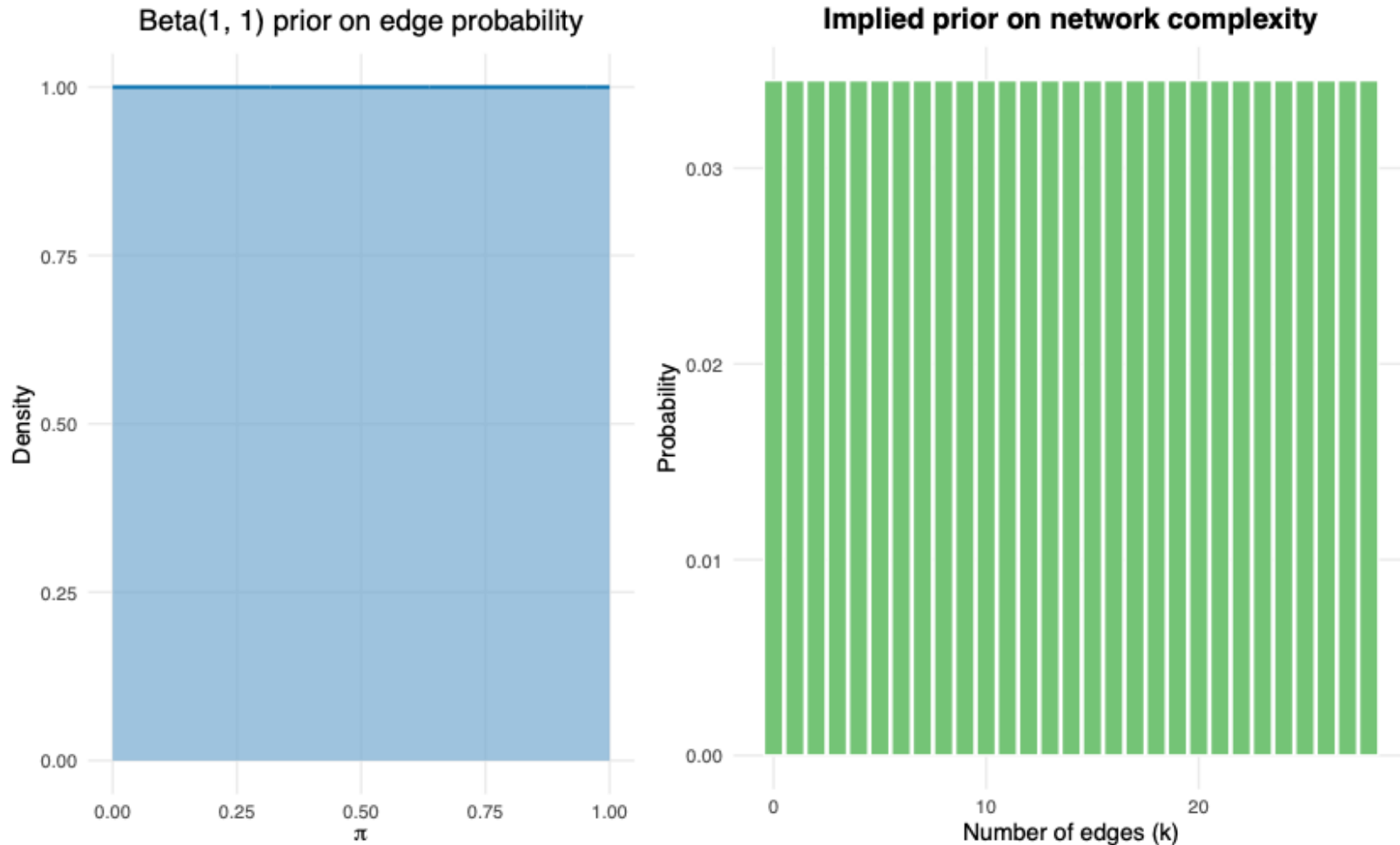
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## The Beta-Bernoulli model

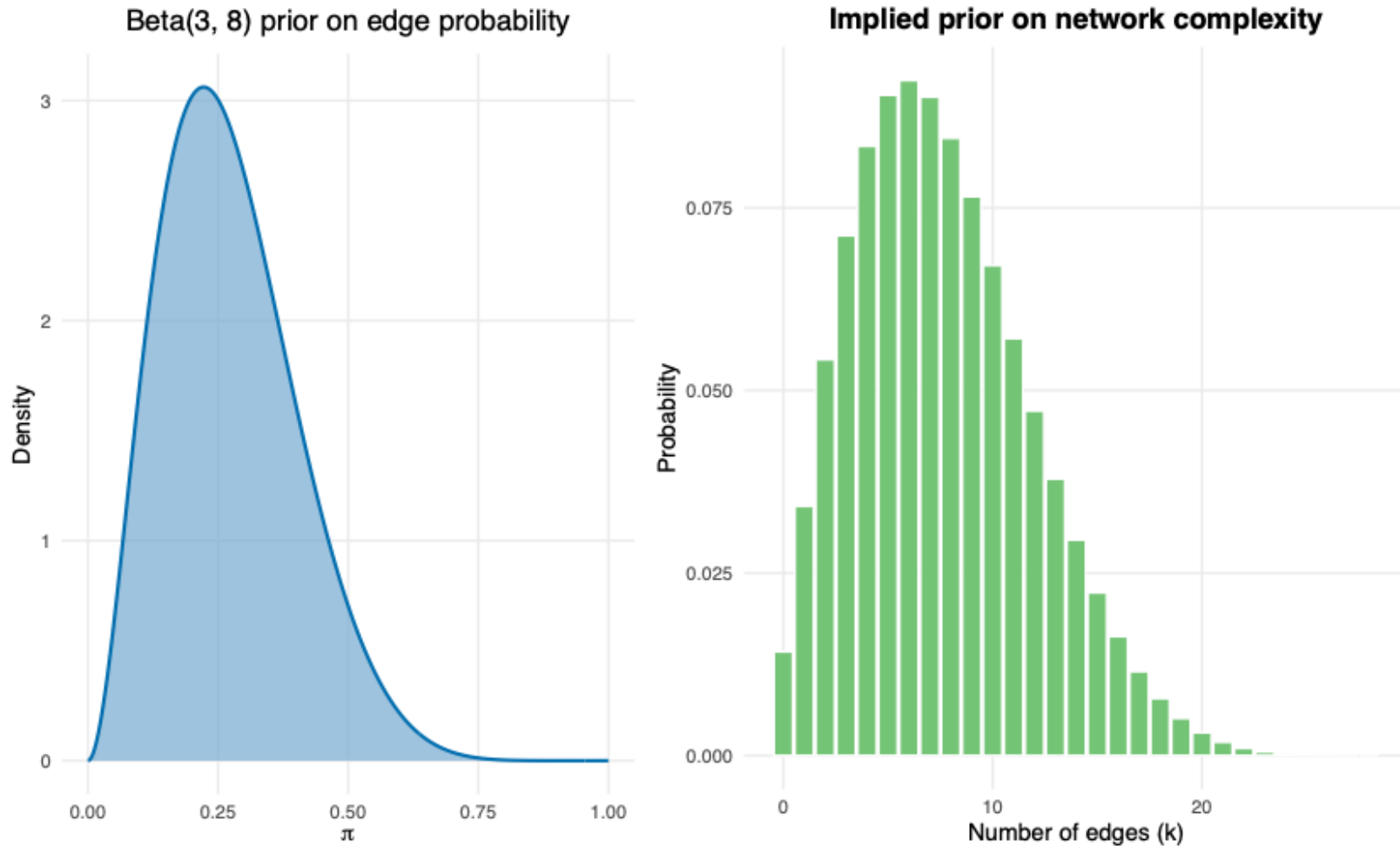
$$\begin{aligned} p(\gamma_{12}, \dots, \gamma_{(p-1)p}) &= \underbrace{\int_0^1 \prod_{i=1}^{p-1} \prod_{j=i+1}^p \pi^{\gamma_{ij}} (1 - \pi)^{1-\gamma_{ij}}}_{\text{Bernoulli model}} \underbrace{\frac{1}{B(\alpha, \beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1} d\pi}_{\text{Beta density}} \\ &= \frac{1}{B(\alpha, \beta)} B(\alpha + k, \beta + m - k) \end{aligned}$$

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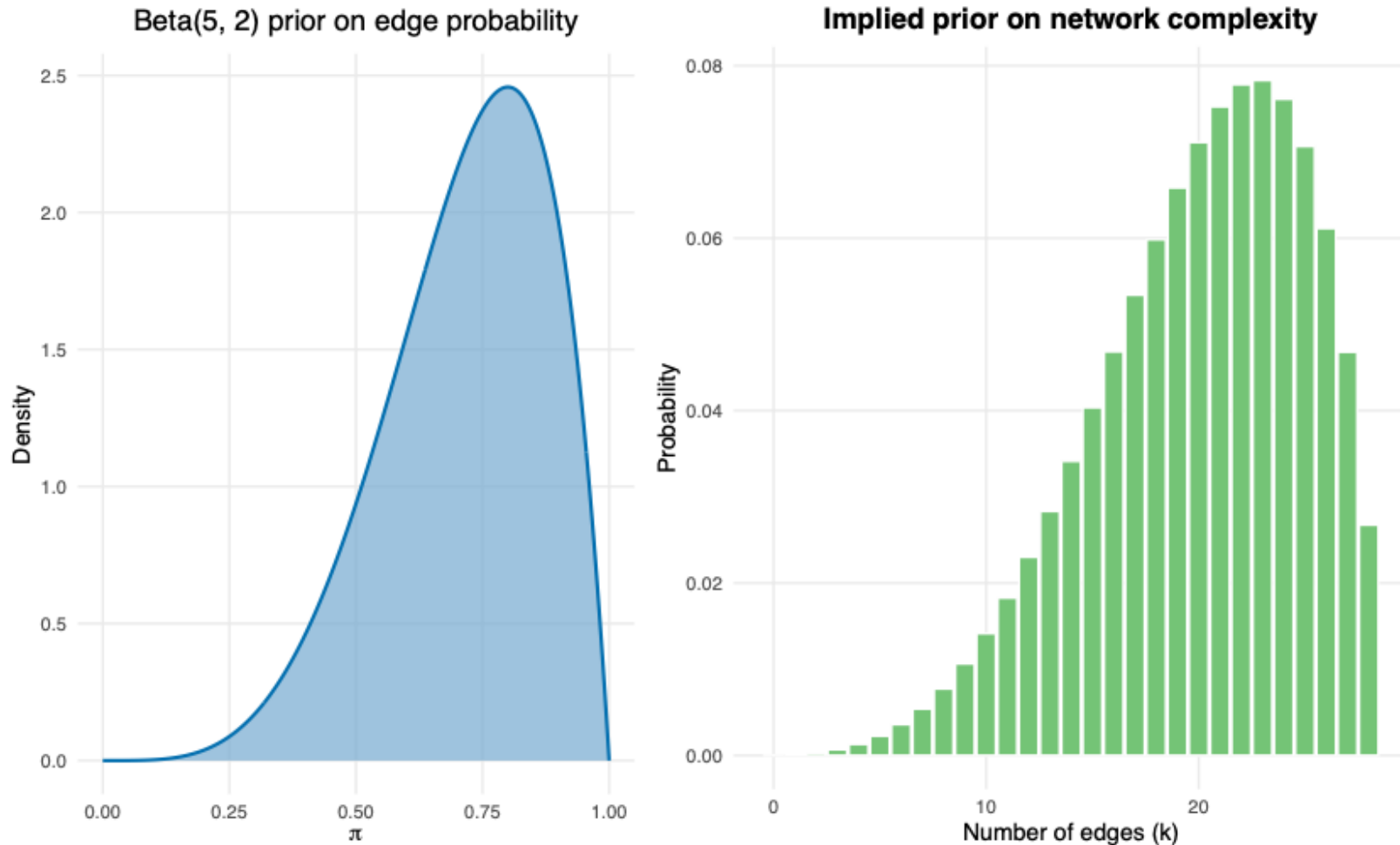
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## The Stochastic-Block model

$$p(\gamma_{12}, \dots, \gamma_{(p-1)p} \mid z_1, \dots, z_p) p(z_1, \dots, z_p \mid b) p(b)$$

Here,  $z$  are cluster allocations,  $m$  total number of clusters (blocks).



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